

COS v3 — Cognitive OS Specification

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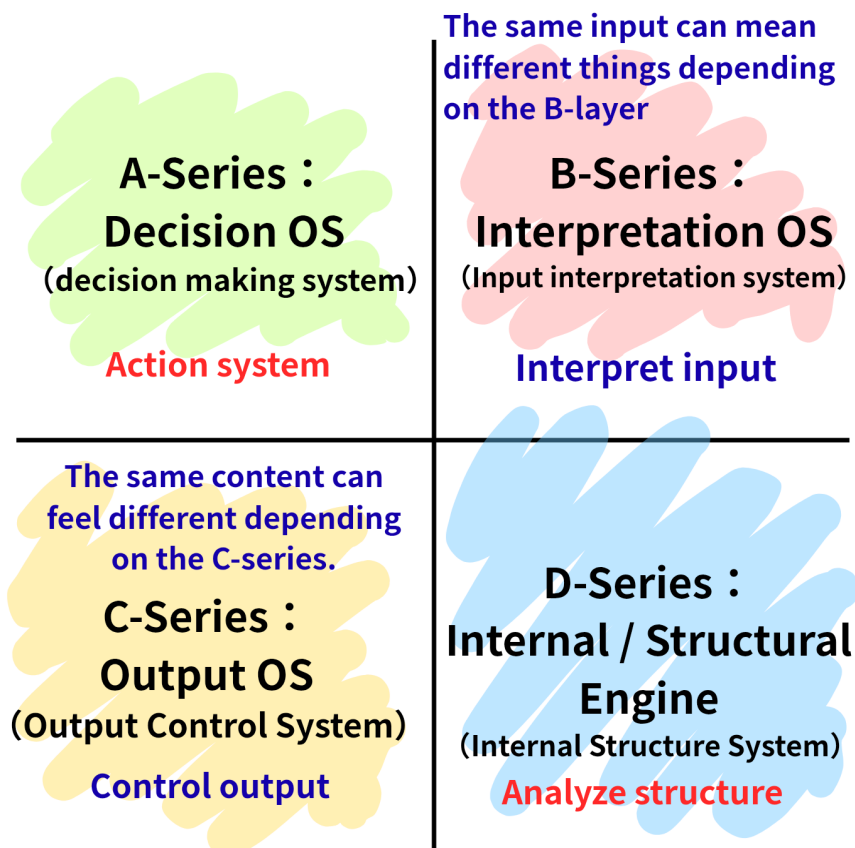
Formal Architecture and Synchronization Protocol

Version 3.0 (Reorganized Edition)

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Scope detects layer errors in thought,
while COS detects misaligned
assumptions. Both operate in parallel.



Abstract

COS v3 is an integrated architecture for modeling human cognitive processing, structured as a three-layer OS model.

COS-Core (A1–D4) — Fundamental cognitive operations

COS-Error — Misinterpretation structures and correction algorithms

COS-Sync — Human–AI synchronization protocol

This document presents COS (Cognitive Operating System) as an executable specification for cognitive processing.

■ Overview of the COS-Core Structure

[1] COS-Core (A1–D4)

[2] COS-Error

[3] COS-Sync

A-Series: Decision OS (Decision Flow)

A1: Premise Recognition

A2: Path Selection

A3: Possibility Expansion

A4: Action Execution

B-Series: Interpretation OS (Reading Flow)

B1: Literal

B2: Context

B3: Intent

C-Series: Output OS (Response Flow)

C1: Explanation

C2: Translation

C3: Dialogue Adjustment

C4: Compression

D-Series: Internal Structural Engine

D1: Human Behavior Architect

D2: Metacognitive Regulator

D3: Self-Structure Researcher

D4: Human–AI Structural Synchronizer

COS

Cognitive Operating System

Scope detects layer errors in thought, while COS detects premise misalignment. Both operate in parallel.

A-Series :

Decision OS

(decision making system)

Action system

A1:

Premise identification

Identify the premises.

A2:

Path selection

Select a path.
Set the goal.

A3:

Scenario exploration

Branch prediction

A4:

Execution

Action planning

B-Series :

Interpretation OS

(Input interpretation system)

Interpret input

B1:

Literal interpretation

Interpret words by their literal meanings.

B2:

Contextual interpretation

Interpret based on context and emotion.

B3:

Intent interpretation

Infer the speaker's intent and goal.

The same input can mean different things depending on the B-layer

C-Series :

Output OS

(Output Control System)

Control output

C1:

Structural explanation

Causal clarification
Logical explanation

C2:

Translation

Make the abstract concrete
Use softer phrasing

C3:

Distance regulation

Reduce pressure
Soften assertions
Prioritize conversation continuity

C4:

Compression

Extreme compression
Assumes shared premises
Structure-focused

The same content can feel different depending on the C-series.

D-Series : Internal / Structural Engine

(Internal Structure System)

Analyze structure

D1:

Action design perspective

Read through Scope
Interpret behavior as structural choice, not emotion

Treat others as structural systems, not personalities.

D2:

Metacognitive control

Detect conversational misalignment
Adjust abstraction level
Re-anchor the premises

Conversation stabilizer

D4:

Human-AI synchronization

Understand AI structurally
Prevent misalignment
Force mode lock
Prevent AI dependency
Final lock layer

D3: Self-structure research

Long-term analysis of thought logs
Not identity-focused
Structural exploration
Self-OS design research

0. Positioning of COS v3

COS v3 defines the human cognitive OS as a three-layer architecture, with A1–D4 placed at its core.

COS v3 =

Core OS (A1–D4)

+ Error Specification

+ Human–AI Synchronization Protocol

In this version, the structural framework of v2 is preserved while reorganizing the document into a formal specification format.

A-Series (Decision Flow)

A1: Premise Recognition

Identify the current situation, available resources, and constraint conditions.

A2: Path Selection

Based on A1, rapidly filter executable strategic routes.

Intuition refers to the high-speed integrated processing of $A1 \rightarrow A2$, where the OS completes the decision before it is verbalized.

A3: Possibility Expansion

For each candidate route, future branches are expanded:

Best-case scenario

Worst-case scenario

Realistic trajectory

A4: Action Execution

Design and implement concrete actions to move toward the selected future trajectory.

B-Series: Interpretation OS (Input Interpretation)

If A1 represents the “premise” and A2 the “structural route,” the B-Series governs how incoming information is interpreted.

B1: Literal Mode

Read the other person’s words in their literal meaning.

B2: Context Mode

Interpret meaning based on the surrounding situation.

B3: Intent Mode

Read what lies behind the statement:

Intent

Background

Underlying motivation

C-Series: Output OS (Output Control)

The C-Series controls the form and style of output.

C1: Explanation Mode

High-logic, high-structure explanation.

C2: Translation Mode

Convert complex or abstract concepts into forms that can be understood.

C3: Relationship / Dialogue Mode

Adjust according to the other person’s state:

Tone

Distance

C4: Compression Mode

Output only the essential points in a highly compressed form.

D-Series: Internal Structural Layer

D1: Human Behavior Architect

Identify the design blueprint underlying human actions and value systems.

D2: Metacognitive Regulator

Instantly detect and correct conversational misalignment through:

Frame alignment
Abstraction adjustment
Premise reconstruction

This functions as a dialogue stabilization mechanism.

D3: Self-Structure Observation

Analyze thought logs over the long term and understand the self not as an identity, but as a structure.

The goal is not self-definition, but the clarification of the cognitive OS structure.

D4: Human–AI Structural Synchronization

Understand systems such as GPT not as black boxes but as structures.
By aligning structures, the AI can be guided into a structural synchronization mode.

This functions as the final safety lock layer of the OS, enabling:

Minimization of the layer gap between AI and humans
Reduction of D2-level misunderstandings
Stabilization of human–AI cooperative processing

The D-Series operates as an oversight layer—similar to a referee or supervisor—continuously monitoring A/B/C processes and intervening when misalignment occurs.